



DETERMINANTS OF VEHICLE (CAR) VALUATION IN THE SRI LANKAN ONLINE SECONDARY MARKET

***A Multi-Dimensional Analysis of the Pre and Early-
2025 Import Liberalization Phases***

**Group 07
ST 3011-Statistical Programming
Department of Statistics
Faculty of Science
University of Colombo**





Meet the Team – Group

07



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INTRODUCTION

- Sri Lanka's vehicle market experienced abnormal pricing during the 2020–2024 import ban, as supply restrictions and inflation caused used vehicles to appreciate rather than depreciate.
- The announcement of import liberalization in early 2025 initiated a speculative correction phase, resulting in unstable and inconsistent market prices.
- In the absence of an official price guide, a growing gap emerged between seller expectations and buyer valuations.
- Currently, no reliable statistical model exists to estimate fair prices in Sri Lanka's digital used vehicle market, as traditional depreciation models fail to reflect evolving consumer preferences.



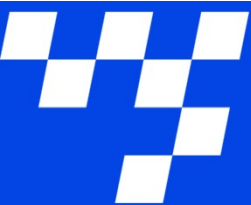


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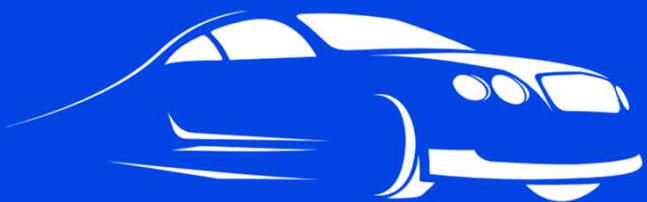
When you are buying a vehicle, actually you're paying for three additional vehicles.

- CEO of Economic Policy Think Tank Advocata





ABOUT THE DATASET



SriLankan Vehicle(Car) Price Dataset

Sri Lanka Used Car Market Dataset: Insights from Ikman.lk & Riyasewana.lk

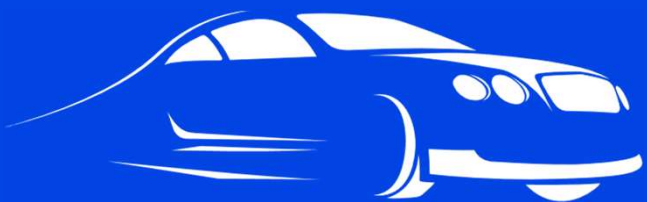
- Insights from [Ikman.lk](#) & [Riyasewana.lk](#), two popular online vehicle marketplaces in Sri Lanka.
- Contains scraped used car listings with key details like price, brand, model, year, mileage, fuel type, and location.
- Dataset Source / Owner: [Prasad Nirmal](#)
- Available on: [Kaggle](#) platform

kaggle





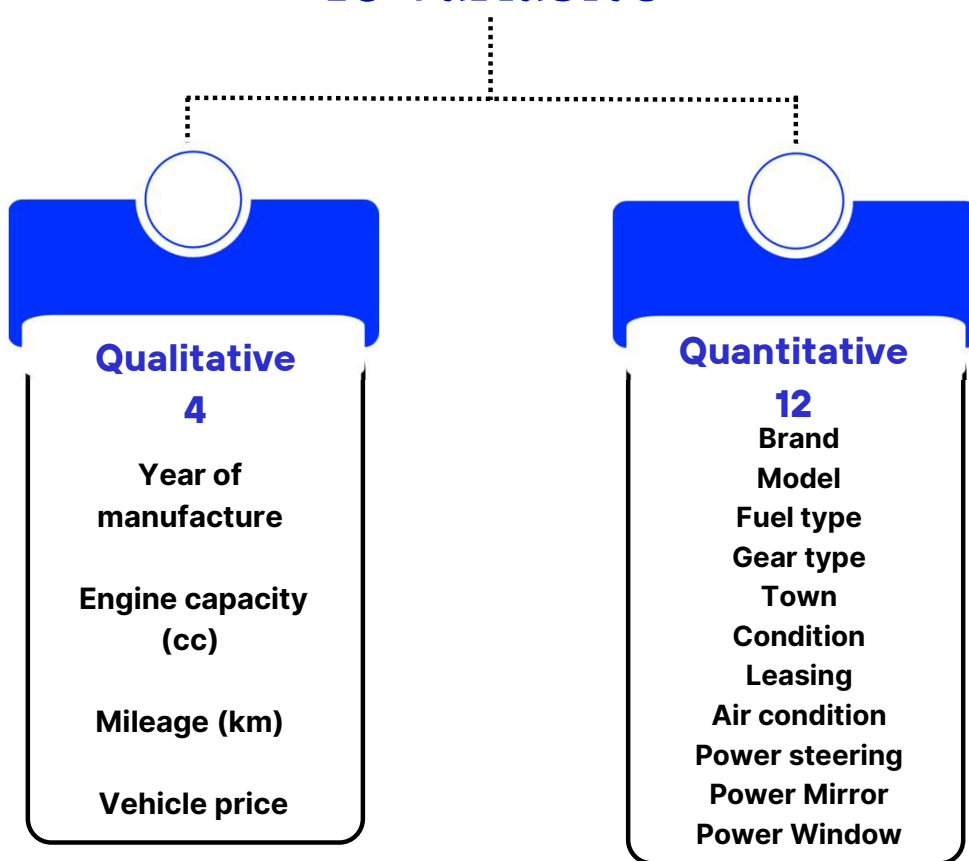
ABOUT THE DATASET



- The dataset contains 9,788 records and 16 variables, of which 12 are categorical while 4 are numerical.
- This is a cross-sectional dataset of vehicle listings from December 3, 2024, to February 5, 2025, capturing pre-liberalization market volatility.



16 Variables



Variable	Definition	Data Type	Details / Range
Brand	The manufacturer of the vehicle	Categorical	50 vehicle brands
Model	The specific model of the car	Categorical	1,555 Models
YOM	Year of Manufacture	Numerical	Discrete (e.g., 2022)
Engine (cc)	Engine size in cubic centimeters	Numerical	Continuous
Gear	Type of transmission	Categorical	Automatic or Manual
Fuel Type	Type of fuel used	Categorical	Diesel, Petrol, Electric, Hybrid
Mileage (KM)	Total distance traveled	Numerical	Continuous
Town	Sale location	Categorical	107 towns
Date	Listing post date	Date	YYYY-MM-DD
Leasing	Leasing options availability	Categorical	Available / Not Available
Condition	Overall status of the car	Categorical	Used or New
Air Condition	Presence of AC	Categorical	Available / Not Available
Power Steering	Presence of power steering	Categorical	Available / Not Available
Power Mirror	Presence of power mirrors	Categorical	Available / Not Available
Power Window	Presence of power windows	Categorical	Available / Not Available
Price	Selling price in lakhs (LKR)	Numerical	Continuous

```

--- Dataframe info ---
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Index: 9788 entries, 0 to 9788
Data columns (total 16 columns):
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 1   Model                 9788 non-null  object  
 2   YOM                   9788 non-null  int64   
 3   Engine (cc)          9788 non-null  float64  
 4   Gear                  9788 non-null  object  
 5   Fuel Type             9788 non-null  object  
 6   Millage(KM)           9788 non-null  float64  
 7   Town                  9788 non-null  object  
 8   Date                  9788 non-null  object  
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11   AIR CONDITION         9788 non-null  object  
12   POWER STEERING        9788 non-null  object  
13   POWER MIRROR          9788 non-null  object  
14   POWER WINDOW          9788 non-null  object  
15   Price                 9788 non-null  float64  
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memory usage: 1.3+ MB
None

--- Number of rows and columns ---
Rows: 9788, Columns: 16

```



PROBLEM DEFINITION AND OBJECTIVES OF THE STUDY



This study aims to identify and quantify the factors associated with second-hand vehicle prices in Sri Lanka during the closed market era (2020–2024), helping stakeholders make informed pricing decisions amid high information asymmetry. Without a standardized valuation system, prices vary widely even for identical vehicles.

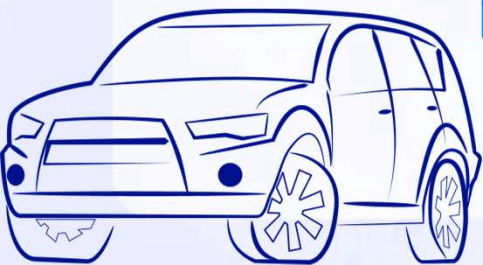
This project addresses the lack of a reliable "Fair Listing Price" model by focusing on four key drivers as follows:

Depreciation (Brand reputation and Age)

Performance (Mechanical specifications like Fuel , Engine size)

Utility (Comfort features and Condition)

Market Dynamics (Location and Financial accessibility)



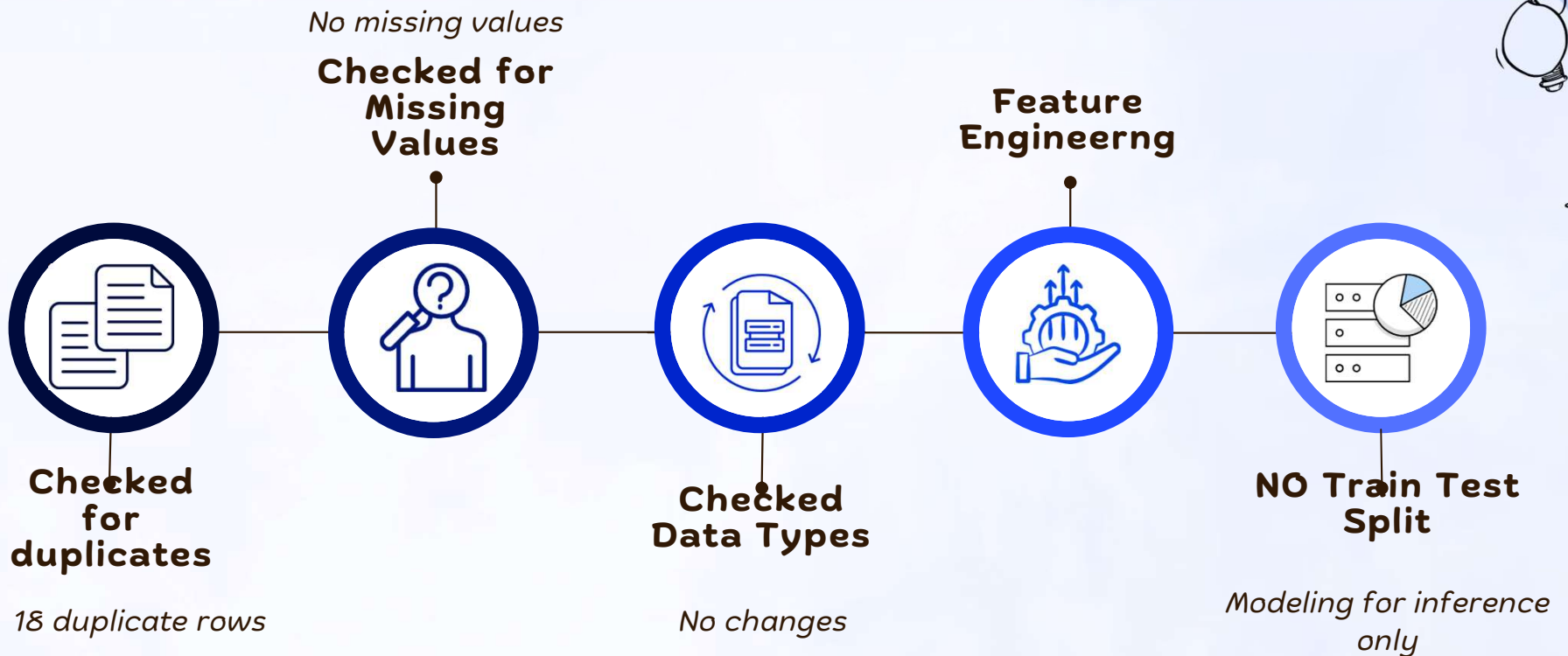
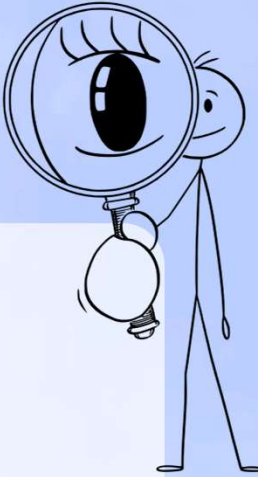


Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

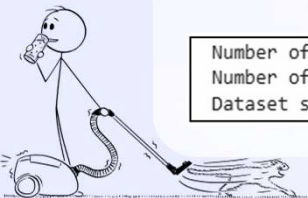
DATA PREPROCESSING



BEFORE EDA



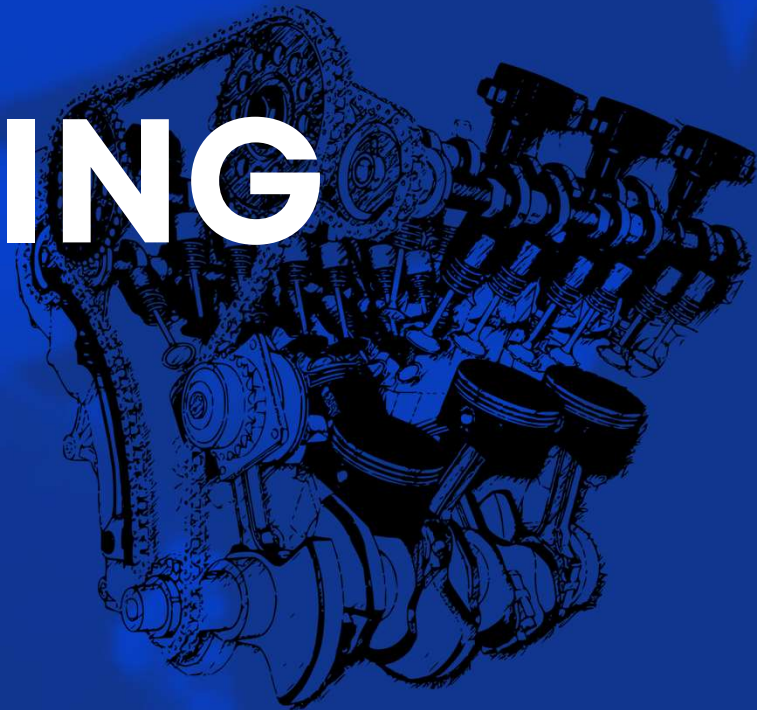
Number of duplicates: 18
Number of missing values: 0
Dataset shape after cleaning: (9770, 16)





Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

FEATURE ENGINEERING



Temporal & Physical Characteristics

(Vehicle) Age



$$\text{Age} = 2025 - \text{YOM}$$

Year Category



$$\text{Year_Category} = \begin{cases} \text{Old} & \text{if YOM} < 2010 \\ \text{Intermediate} & \text{if } 2010 \leq \text{YOM} \leq 2017 \\ \text{Modern} & \text{if YOM} \geq 2018 \end{cases}$$

Engine Segment



$$\text{Engine_Segment} = \begin{cases} \text{Micro} & \text{if cc} < 800 \\ \text{Compact} & \text{if } 800 \leq \text{cc} \leq 1200 \\ \text{Mid-Range} & \text{if } 1200 < \text{cc} \leq 1600 \\ \text{Large} & \text{if cc} > 1600 \end{cases}$$

Leasing Status & Geographical Characteristics

Leasing (Encoding)



Ongoing Leasing → True
No Leasing → False

Province



Colombo → Western
Gampaha → Western
Kandy → Central
Galle → Southern

Location Type



Municipal / Urban Council = Yes → Urban
Municipal / Urban Council = No → Non-Urban

Brand & Model Optimization

Brand Grouped



$$\text{Brand_Grouped} = \begin{cases} \text{Original Name} & \text{if } n > 100 \\ \text{OTHER} & \text{if } 10 \leq n \leq 100 \\ \text{RARE} & \text{if } n < 10 \end{cases}$$

Simple Model



A4 RS4 CONVERSION → A4
A4 S LINE → A4
A6 S LINE → A6

Brand Model



Brand Grouped = Honda
+
Simple Model = FD1 } Brand Model = Honda_FD1

EXPLORATORY DATA ANALYSIS (EDA) OF THE STUDY



Distribution of Vehicle Prices

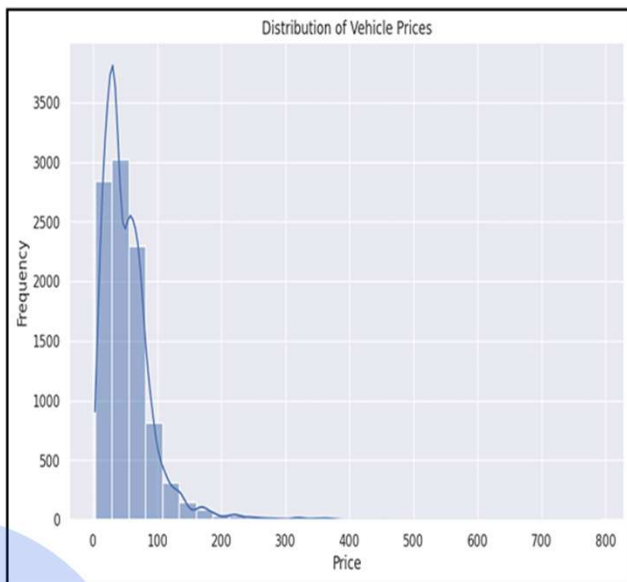


Figure 3: Distribution of Price

- The distribution of vehicle prices is highly right-skewed, with most listings concentrated in lower price ranges. A long right tail indicates the presence of a small number of high-value vehicles.
- The price distribution was examined for bimodality, which could indicate multiple market segments.
- The Bimodal Coefficient is below the threshold of 0.555, suggesting no clear bimodal structure.
- Ashman's D indicates poor separation between potential modes.
- Hartigan's Dip Test rejects strict unimodality, suggesting some departure from a single peak.
- Overall, the results indicate that the price distribution is not strongly bimodal, but rather complex and continuous.

Distributional Characteristics of Numerical and Categorical Variables

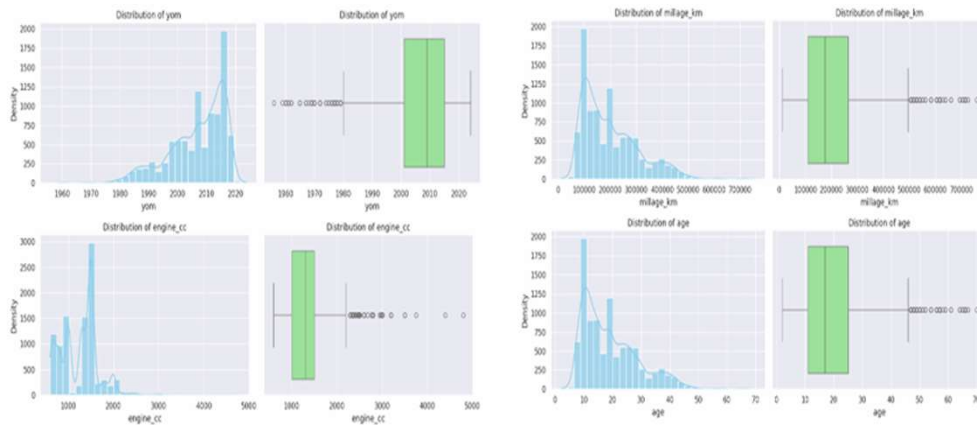


Figure 4 : Distribution of numerical variables

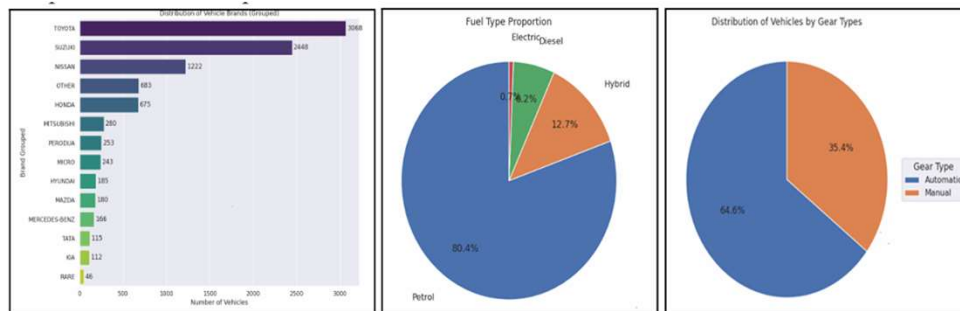


Figure 5 :Distribution of vehicles by brand, fuel type, and gear type.

- variables such as price, mileage, and vehicle age exhibiting strong positive skewness.
- These characteristics motivate the use of robust and non-parametric methods in later analysis.
- The categorical distributions show that the vehicle market is highly concentrated by brand, with Toyota and Suzuki dominating the listings.
- Petrol vehicles account for the majority of listings, while hybrid and diesel vehicles form smaller segments and electric vehicles remain negligible.
- Automatic transmission vehicles dominate the market, suggesting a preference for driving convenience.
- Together, these numerical and categorical patterns highlight the heterogeneous and uneven structure of the second-hand vehicle market, justifying the use of robust analytical methods in later stages.

Outlier Identification and Treatment



Feature	Q1	Q3	IQR	Lower_Bound	Upper_Bound	Outlier_Count	Outlier_Percentage
price	26.3375	69.7625	43.425	-38.8	134.9	381	3.937578
engine_cc	1000.0000	1500.0000	500.000	250.0	2250.0	109	1.126499
yom	2001.0000	2015.0000	14.000	1980.0	2036.0	75	0.775114
millage_km	110000.0000	264000.0000	154000.000	-121000.0	495000.0	75	0.775114
age	11.0000	25.0000	14.000	-10.0	46.0	75	0.775114

Figure 6: Outlier detection for numerical variables.

- Outliers were identified using the Interquartile Range (IQR) method.
- Most numerical variables contain fewer than 1% outliers, except price.
- These extreme values were retained, as they represent valid market observations such as luxury or rare vehicles.
- Removing these observations could result in loss of important market information.

Correlation Between Numerical Variables and Multicollinearity Assessment

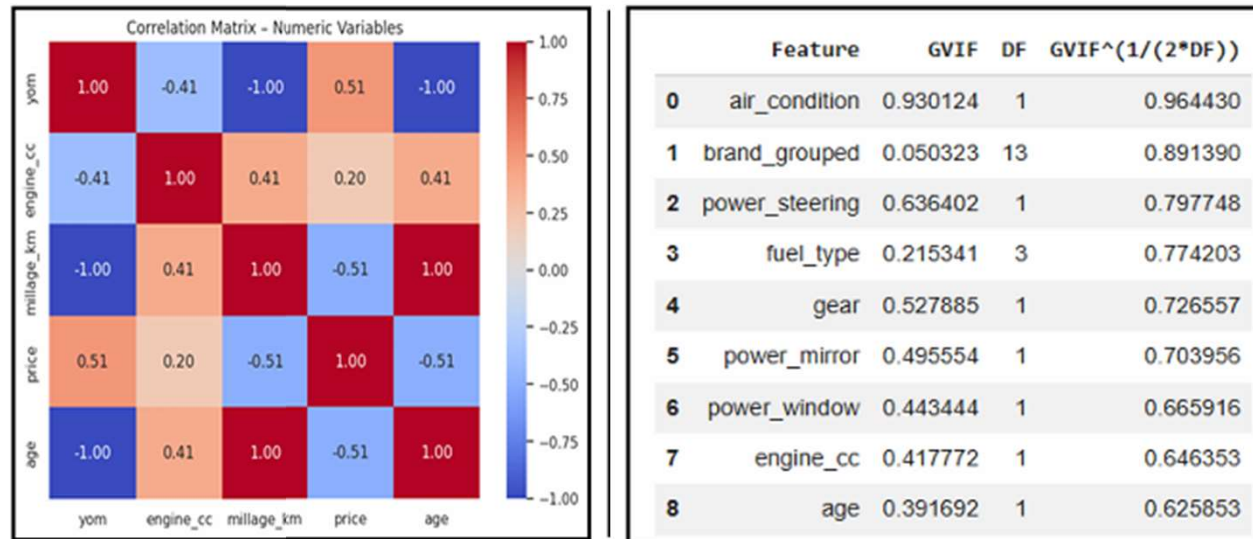


Figure 7: Correlation matrix and GVIF values for explanatory variables.

- The correlation matrix shows a strong positive relationship between vehicle age and mileage.
- Vehicle price is positively correlated with Year of Manufacture and negatively correlated with age and mileage.
- These relationships are consistent with expected depreciation patterns in the vehicle market.
- Generalized Variance Inflation Factors (GVIF) were computed to assess multicollinearity.
- All variables exhibit low adjusted GVIF values.
- This indicates that the explanatory variables are sufficiently independent.



Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

OBJECTIVE 01:

Brand Equity and Depreciation Analysis



WHY?



The Valuation Challenge:

Determining the fair market value of a second-hand vehicle is complex because price isn't just about age. It's a delicate balance of brand prestige, mechanical wear, and technical specifications.

The Goal:

To move beyond "gut feeling" and quantify exactly how much a brand name is worth and how quickly that value disappears over time, ensuring pricing strategies are data-driven rather than anecdotal.



Quantifying the "Brand Premium"

- Isolating the monetary value added by a brand name alone.
- Comparing luxury vs. economy brands while holding engine size and age constant.

Non-Linear Depreciation Mapping

- Moving beyond simple linear "loss per year."
- Identifying the "cliff" (years where value drops fastest) and the "plateau" (where value stabilizes).

Statistical Framework

- Why OLS Failed: Standard linear models couldn't capture the complex, curving nature of vehicle aging.
- The GAM Approach: Utilizing Generalized Additive Models to accurately map non-linear patterns while maintaining statistical control.

Data Refinement & Logic



Solving Multi-Collinearity (Age vs. Mileage)

- **The Problem:** Vehicle Age and Mileage were found to be perfectly correlated, creating redundant data noise.
- **The Experiment:** Attempted a "Residualization Approach" to isolate unique mileage data.
- **The Decision:** Since residualization showed no statistical improvement, Mileage was removed.
- **Result:** Age is retained as the cleaner, primary driver of depreciation.

Managing High Cardinality (Model vs. Engine Size)

- **The Problem:** Including every specific vehicle model would create too many variables (High Dimensional Noise), making the model uninterpretable.
- **The Solution:** Excluded specific "Models" and used Engine Capacity engine_cc as a proxy.
- **The Benefit:** engine_cc acts as a continuous variable that captures performance and market positioning without the complexity of hundreds of model names.



Selected Variables for Exploratory Analysis



Variable	Represent
Vehicle Age	Physical
Engine Capacity	Perform

Strategic Goal:

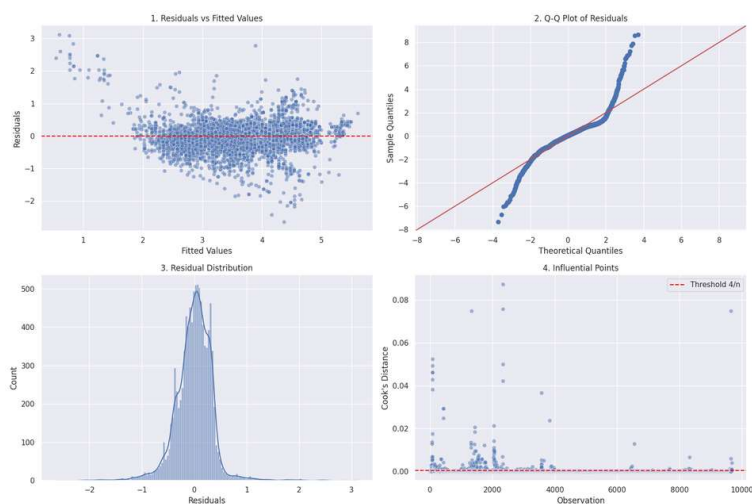
Exploratory over Predictive The model is designed to explain relationships (why prices change) rather than just blindly predicting a number. By keeping the model lean, we ensure that every result is statistically significant and easy to translate into business insights.



Testing the Linear Model Assumptions



Diagnostic Plots for Engine-Adjusted Brand Model



The Conflict:

Initial visual inspections were promising, but rigorous statistical tests revealed hidden flaws in the data structure.

The Goal:

To determine if a standard Ordinary Least Squares (OLS) model could provide valid coefficients for hypothesis testing.

Test Metric	Visual Observation (Initial)
Normality	Histogram: Suggested
Variance	Residuals vs. Fitted: A homoscedastic (even :

Addressing Model Invalidity



Why we cannot use OLS Coefficients:

- The contradicting diagnostic results mean OLS coefficients are not valid measurements for formal hypothesis testing.
- Relying on them would lead to "Type I" or "Type II" errors (false positives/negatives) in our brand analysis.

The Strategic Shift:

- Methodology: Switched to Non-Parametric Methods.
- Advantage: These methods do not rely on restrictive assumptions like normality or constant variance.
- Outcome: Provides a mathematically "honest" foundation for quantifying brand equity and depreciation without being skewed by data outliers or non-linear noise.

Statistical Validation of Brand Influence



The Kruskal-Wallis Test

- Objective: To determine if price distributions differ across all car brands (using log-transformed prices to handle skewness).
- The Result: **H-Statistic: 2682.46**
 - **P-value: $p < 0.05$**
- Conclusion: We Reject the Null Hypothesis. There is overwhelming evidence that car prices are not distributed equally across brands.

Pairwise Refinement (Mann-Whitney U)

- Used Bonferroni correction to prevent "false discovery" (Type I errors) during multiple comparisons.
- Finding: Almost all brands show a statistically distinct price distribution compared to Toyota, confirming that Toyota occupies a unique market "niche."

Directional Price Analysis: The Toyota Benchmark



Through one-sided Mann-Whitney U tests, we identified Toyota's specific "Price Tier" in the Sri Lankan market:

Comparison Group	Brand I
Luxury / Premium	Mercedes
Budget / Entry Level	Nissan

Takeaway:

Toyota acts as the "Middle-Market Anchor." It commands a premium over other Japanese and Indian manufacturers but remains accessible compared to European luxury brands.

Partial Correlation Analysis: Age and Engine Capacity



By using Spearman's Partial Correlation, we isolated the effect of each variable while controlling for the others.

The Impact of Engine Capacity (engine_cc)

- Correlation ρ : +0.370 (Moderate Positive)
- Insight: Even when two cars are the same age and brand, a larger engine consistently leads to a higher price. This confirms engine_cc as a valid proxy for performance and luxury.

The Impact of Vehicle Age

- Correlation ρ : -0.814 (Strong Negative)
- Insight: Age is the dominant driver of price. The extremely high negative correlation shows that depreciation is a relentless force that overrides almost all other factors.

Modeling Non-Linear Market Dynamics



Model Framework:

- **Target Variable: log(price)** — Log-transformed to stabilize variance and handle right-skewness.
- The Predictors:
 - Brand (Categorical): Toyota selected as the Reference Brand (The "Baseline") due to its high market share and representative positioning in Sri Lanka.
 - Age (Continuous): Modeled with a smoothing function to capture non-linear depreciation.
 - log(engine_cc) (Continuous): Log-transformed to account for diminishing returns in performance-based pricing.

LinearGAM					
=====					
Distribution:	NormalDist	Effective DoF:		43.1395	
Link Function:	IdentityLink	Log Likelihood:		-7413.1481	
Number of Samples:	9676	AIC:		14914.5752	
		AICc:		14914.989	
		GCV:		0.2744	
		Scale:		0.5217	
		Pseudo R-Squared:		0.5112	
=====					
Feature Function	Lambda	Rank	EDoF	P > x	Sig. Code
=====					
s(0)	[0.6]	20	14.2	1.11e-16	***
f(2)	[0.6]	14	12.9	1.11e-16	***
s(1)	[0.6]	20	16.0	1.11e-16	***
intercept		1	0.0	1.11e-16	***
=====					
Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1					

Modeling Non-Linear Market Dynamics



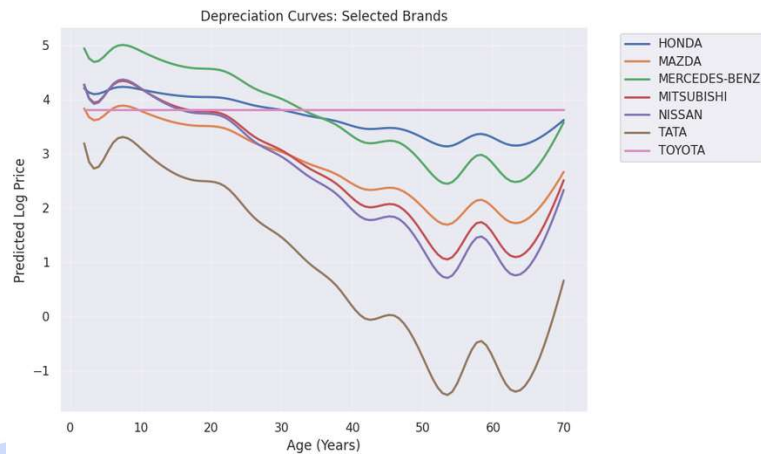
Why This Setup?

- Toyota as the Baseline: By setting Toyota as the intercept, the coefficients for other brands (Mercedes, Suzuki, etc.) directly represent their Price Premium or Discount relative to a standard Toyota.
- Log Relationship: Applying logs to both Price and Engine Capacity allows us to interpret the relationship as Elasticity—how a percentage change in engine size impacts the percentage change in price.

How Brands Lose Value: The Toyota Benchmark



In Sri Lanka's market, vehicle scarcity and import restrictions have turned Toyota into the ultimate "Value Anchor." All other brands are measured by how much faster they fall compared to this baseline.



The Gold Standard (Toyota & Honda):

Toyota: Maintains the shallowest depreciation curve due to high demand and perceived reliability.

Honda: Closely tracks Toyota, signaling a strong local preference for Japanese engineering over European alternatives.

The "Semi-Luxury" Resilient (Mazda):

While it depreciates faster than Toyota, Mazda outperforms Nissan and Mitsubishi, benefiting from its "premium-tier" branding.

The Luxury Decline (Mercedes-Benz):

Faces steep relative depreciation. High maintenance costs and the "status premium" of new models make aging luxury assets less desirable.

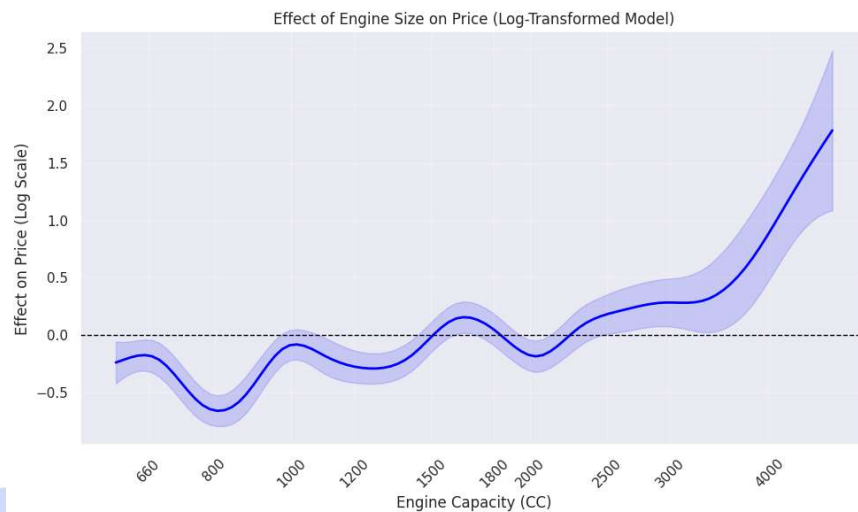
The Budget Floor (Tata):

Exhibits the most severe value loss, as buyers in the budget tier prioritize low initial cost over long-term resale value.

Beyond the Piston: engine_cc and Market Tiering



While there is a clear positive correlation between engine size and price, the data reveals that engine capacity is a proxy for the entire vehicle segment.



Under 1500cc (The Budget/Utility Tier):

Characterized by high price sensitivity. Smaller engines represent economy models where functional utility is the primary value driver.

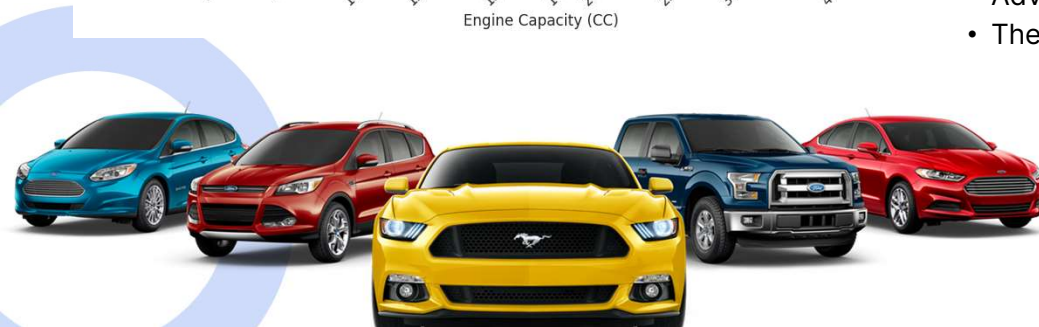
1500cc – 2000cc+ (The Premium/Luxury Transition):

This is the "Inflection Point." Crossing into this capacity usually moves the vehicle from a basic commuter to a Sedan or Luxury SUV.

The "Compound Value" Effect:

The upward price curve isn't just charging for "more power." It captures the bundled value of:

- Superior build quality and materials.
- Advanced technology and safety suites.
- The social status associated with larger, high-performance vehicles.





Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

OBJECTIVE 02:



Mechanical Performance

Analysis



OVERVIEW



- The purpose of this objective is to understand how key mechanical characteristics of cars affect their market price in Sri Lanka's second-hand vehicle market.
- This objective focuses on examining whether mechanical specifications affect vehicle prices.
- Analyzing mechanical specifications is important because it provides insights into which features buyers value most, helps sellers price vehicles fairly, and increases market transparency by reducing overpricing and misinformation.



Fuel-efficient technologies

How Fuel type affects price and whether Hybrid and Electric vehicles command higher prices than traditional Petrol and Diesel cars.



Transmission type

How transmission type affects price and whether Automatic vehicles are priced higher than Manual vehicles due to driving convenience.



Engine capacity (cc)

How engine size affects price and whether it modifies the impact of fuel type or transmission on price.

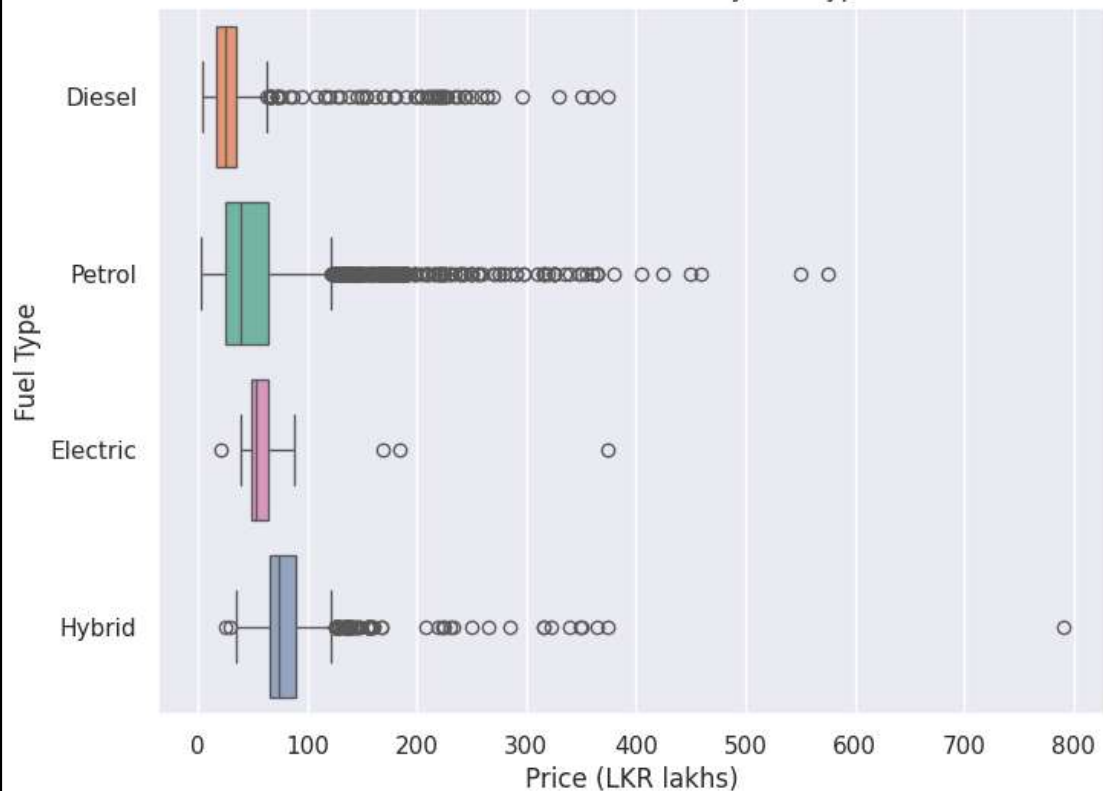
OBJECTIVE 02

EDA / BIVARIATE ANALYSIS

Car Prices by Fuel Type Hybrids Lead the Market



Distribution of Car Prices by Fuel Type



- Hybrid cars have the highest median price and the widest price range, including an extreme value near 790 LKR lakhs which is unusually expensive compared to most Hybrid cars.
- All fuel types have some high-end outliers. There's a very expensive Hybrid car around 800 LKR lakhs. Most fuel types are right-skewed, meaning a majority of cars are in the lower price ranges, with fewer extremely expensive cars.

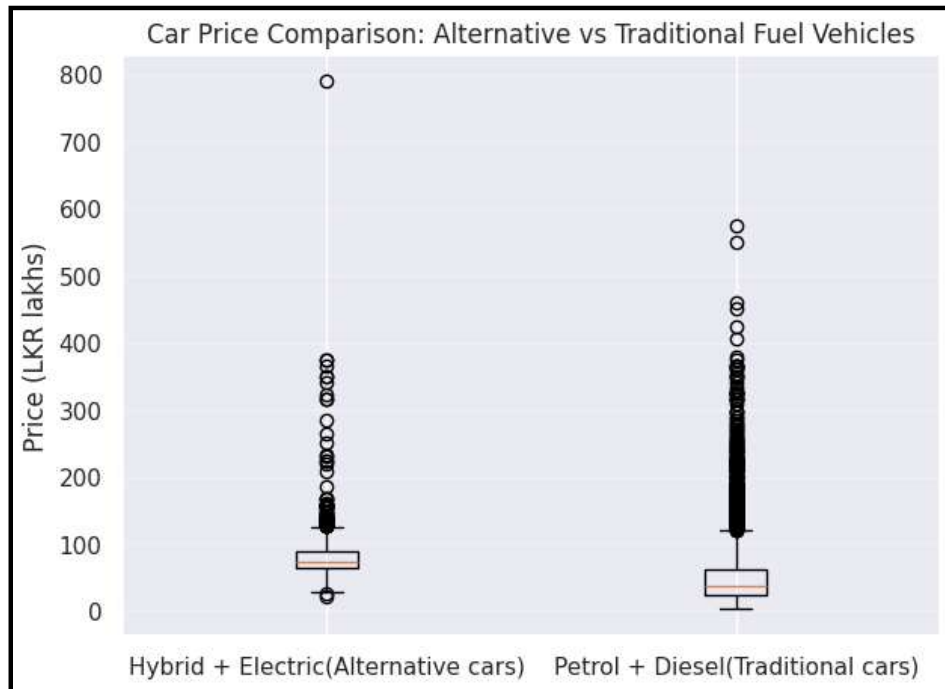


OBJECTIVE 02

EDA / BIVARIATE ANALYSIS

Price Distribution: Traditional vs. Alternative Fuel Cars

Alternative cars Lead the Market



- Traditional (Petrol & Diesel) vs. Alternative (Hybrid & Electric) to show market differences.
- Alternative cars have higher median prices and a wider upper range; both groups show right-skewed distributions.
- Toyota LEXUS 600HL priced at 790 lakhs LKR, far above other cars in the category.
- High prices reflect market distortions during the closed period, not typical conditions.

OBJECTIVE 02

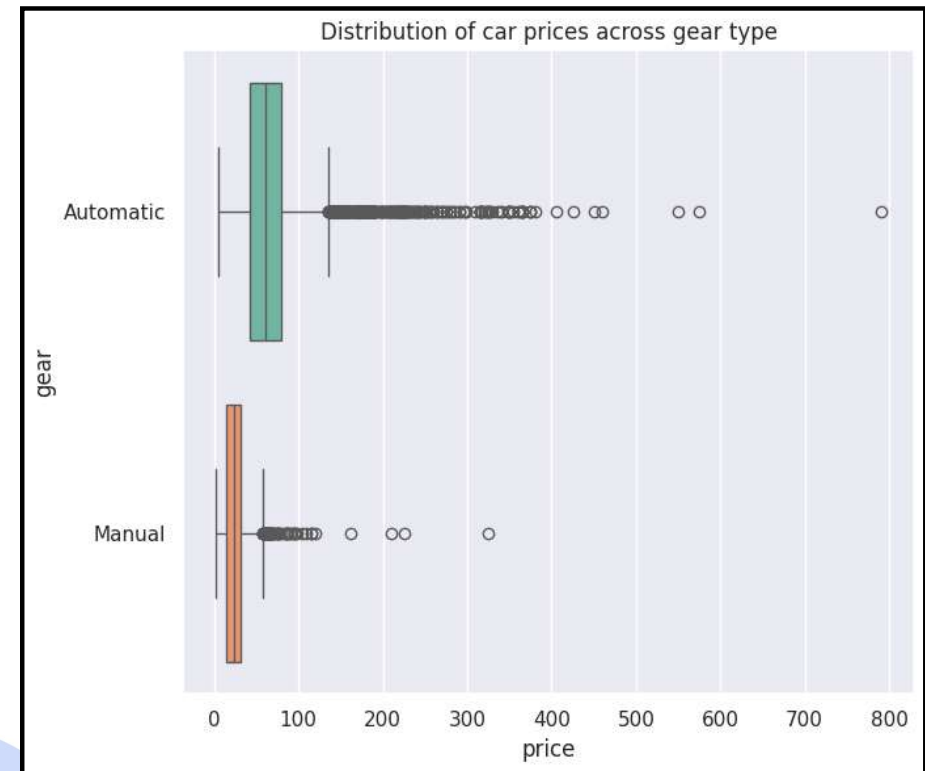
EDA / BIVARIATE ANALYSIS

Price Comparison: Automatic vs. Manual Cars

Automatic Cars Lead the Market



- Automatic cars have higher median prices than manual cars.
- Both types are right-skewed, but automatic cars show more extreme high prices.
- Manual cars are generally lower-priced with fewer outliers, while automatic cars dominate the high-price segment.



OBJECTIVE 02

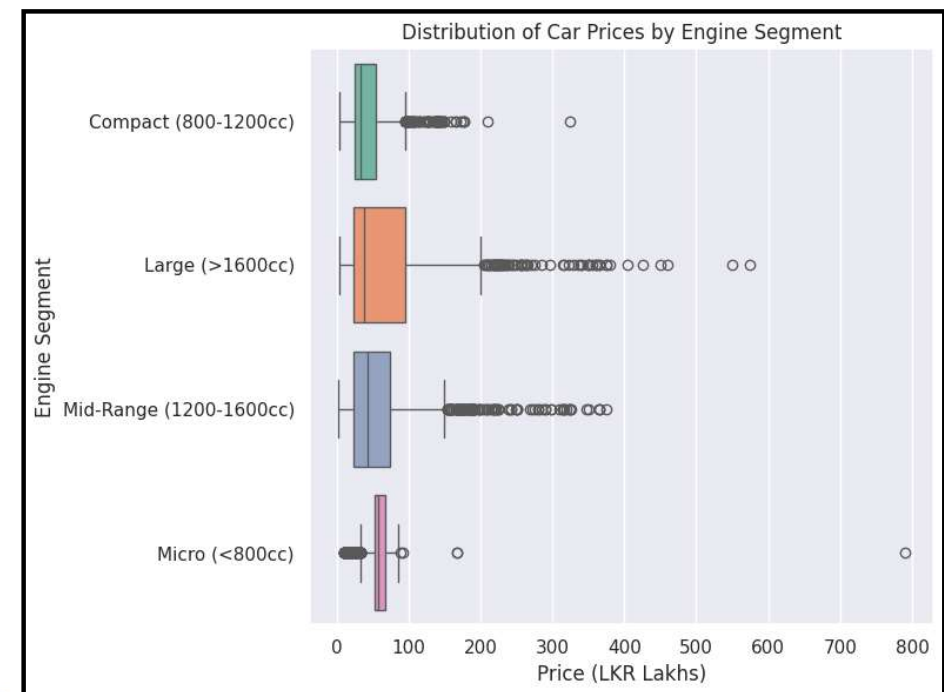
EDA / BIVARIATE ANALYSIS

Car Price Distribution Across Engine Capacity Segments

Micro Engine Capacity Cars Lead the Market



- All segments show right-skewed distributions with long tails due to high-value vehicles.
- Micro segment has a slightly higher median than larger segments.
- Large segment shows the greatest variability and most extreme high prices, despite a lower median.

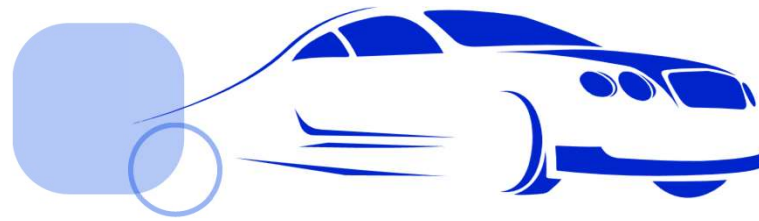


OBJECTIVE 02

ADVANCED ANALYSIS

TESTING PRICE VARIATION ACROSS CAR CATEGORIES

To understand the relationships between car prices and key mechanical characteristics, advanced statistical methods were applied to answer the following research questions.



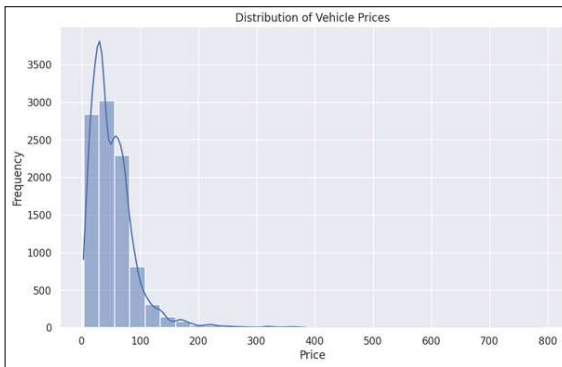
- 01 Do car prices differ across fuel types (Petrol, Diesel, Hybrid, Electric)?**
- 02 Do Alternative fuel cars (Hybrid and Electric) command higher prices than Traditional fuel cars (Petrol and Diesel) in the second hand car market?**
- 03 How transmission(gear) type affects price and whether Automatic cars are priced higher than Manual cars due to driving convenience?**
- 04 Do Vehicle Prices Differ Across Engine Types?**

PARAMETRIC TEST REQUIREMENTS

NORMALITY & VARIANCE

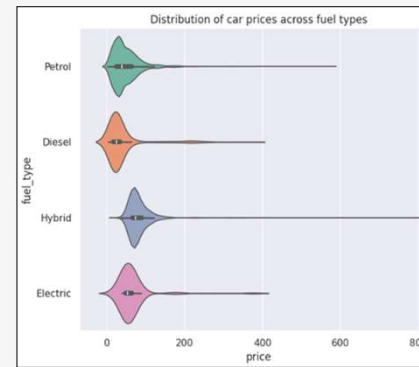


Car Price



- car price is not normal distributed.

Distribution of car prices across fuel types



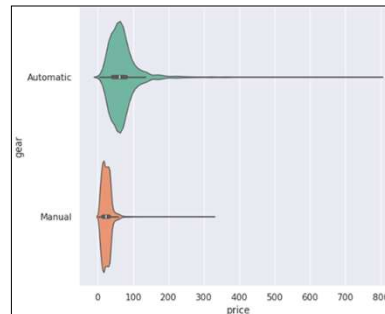
levene_stat=15.39
levene p value =0.00
Decision: Reject the null hypothesis; variances are not equal

- The distribution of car prices within each fuel type group was found to deviate from normality
- Levene's test results showed that the variances were not equal across the groups

Distribution of car prices across gear type

Auto: W=0.732, p=0.0000
Manual: W=0.759, p=0.0000
Auto group is NOT normally distributed
Manual group is NOT normally distributed

- Shapiro wilk test results showed that the distribution of car prices within each gear type group was found to deviate from normality



Do Not Meet Parametric Test Assumptions

Since the violations of normality and homoscedasticity of car prices across fuel type, gear type and engine segments, **non parametric tests** such as **Kruskal-Wallis**, **Mann-Whitney tests** and **post hoc (Dunn's test with Bonferroni correction and Holm correction)** tests were used to test the hypotheses.

DO CAR PRICES DIFFER ACROSS FUEL TYPES (PETROL, DIESEL, HYBRID, ELECTRIC)?



Kruskal Wallis test was applied

H_0 : Median vehicle prices are equal across fuel types (Petrol, Diesel, Hybrid, Electric).

H_1 : At least one fuel type has a different median price.

Test Results

```
Kruskal-Wallis statistic = 1606.2235
Kruskal-Wallis p-value   = 0.0000
Decision: Reject  $H_0$  : median prices differ across fuel types
```

The test provided sufficient evidence of a **significant difference in median prices across fuel types** ($H = 1606.22, p = 0.00$) at 5% level of significance.

Post hoc test (Dunn's test) was used to identify which fuel type pairs differ.

Post hoc test (Dunn's test) Results

Comparison	Raw p-value	Adjusted p-value	Significant?
Petrol vs Diesel	0.0000	0.0000	Yes
Petrol vs Hybrid	0.0000	0.0000	Yes
Petrol vs Electric	0.0001	0.0006	Yes
Diesel vs Hybrid	0.0000	0.0000	Yes
Diesel vs Electric	0.0000	0.0000	Yes
Hybrid vs Electric	0.0001	0.0003	Yes

All pairwise comparisons were statistically significant at 5% significance level.

That means the median prices of all fuel types differ from each other at 5% significance level.

DO ALTERNATIVE FUEL CARS (HYBRID AND ELECTRIC) COMMAND HIGHER PRICES THAN TRADITIONAL FUEL CARS (PETROL AND DIESEL) IN THE SECOND HAND CAR MARKET?



**Mann Whitney U
(one sided)
test was used .**

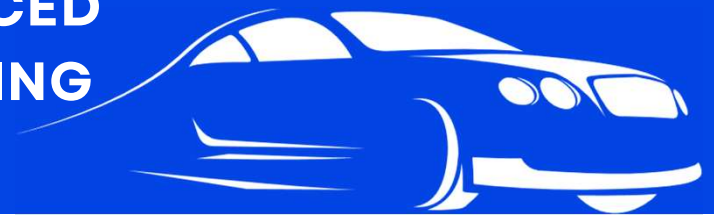
H₀:Median prices of alternative fuel cars are equal to those of traditional fuel cars.

H₁:Median prices of alternative fuel cars are higher than those of traditional fuel cars.

Test Results

- The test provided a p value effectively equal to zero (6.65×10^{-309}) far below the 5% significance level.
- This provides strong evidence to reject H₀ indicating that **alternative fuel cars have higher prices reflecting a clear market premium for hybrid and electric cars.**

HOW TRANSMISSION(GEAR) TYPE AFFECTS PRICE AND WHETHER AUTOMATIC CARS ARE PRICED HIGHER THAN MANUAL CARS DUE TO DRIVING CONVENIENCE?



**Mann Whitney U
(one sided)
test was used .**

**H₀: Median prices of Automatic
and Manual vehicles are equal.**

**H₁: Automatic vehicles have
higher median prices than
Manual vehicles.**

Test Results

- The test yielded a Mann Whitney U statistic of 19,787,379.5 with a p-value =0.00 .
- At the 5% significance level, this provides strong evidence to **reject H₀ indicating that automatic vehicles are priced higher than manual cars .**
- This result suggests that transmission(gear) type is associated with the car price because it reflects driving convenience. Automatic cars are easier to drive in traffic areas and are simpler for new drivers to use compared to manual cars.

DO VEHICLE PRICES DIFFER ACROSS ENGINE TYPES?



Kruskal Wallis test was applied

H₀: The median vehicle price is the same across all engine types.

H₁: At least one engine type has a different median vehicle price.

Test Results

Kruskal-Wallis H=340.979, p=0.000

The test provided sufficient evidence of a **significant difference in median prices across engine segments**. (H = 340.979, p =0.00) at 5% level of significance.

Post hoc test (Dunn's test with Bonferroni correction) was used to identify which engine segment pairs differ

Post hoc test (Dunn's test) Results

Comparison	p-value	Significant?
Compact (800-1200cc) vs Mid-Range (1200-1600cc)	1.932567e-29	Yes
Compact (800-1200cc) vs Large (>1600cc)	1.716977e-19	Yes
Compact (800-1200cc) vs Micro (<800cc)	1.701413e-70	Yes
Mid-Range (1200-1600cc) vs Large (>1600cc)	3.502945e-01	No
Mid-Range (1200-1600cc) vs Micro (<800cc)	2.122961e-25	Yes
Large (>1600cc) vs Micro (<800cc)	1.283856e-09	Yes

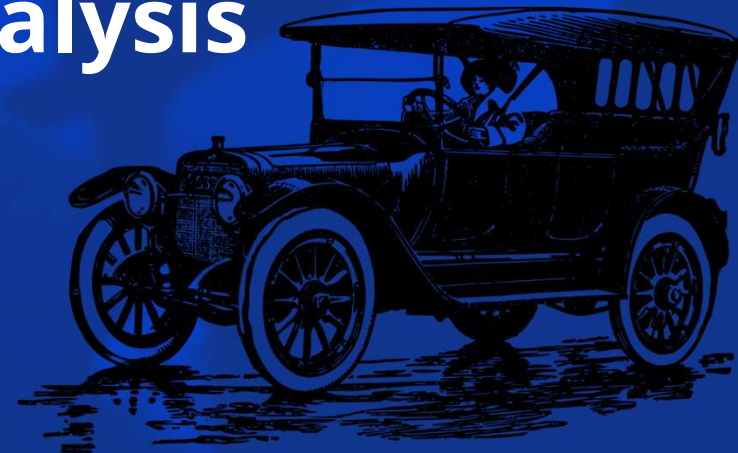
Mid-range (1200–1600cc) and Large (>1600cc) vehicles do not show a statistically significant difference in median prices, while all other engine-type pairs differ significantly at 5% level of significance.



Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

OBJECTIVE 03:

Comfort Feature Analysis



OVERVIEW

- Evaluates how essential comfort features such as Air Conditioning, Power Steering, Power Mirrors, and Power Windows influence vehicle prices in the Sri Lankan secondary market .
- Utilizes visual trend analysis to identify market saturation and non-parametric statistical testing to quantify the price premiums associated with these features.
- Identifies whether these features remain "premium add-ons" or have transitioned into "basic expectations" and provides understanding of how feature rich vehicles resisted standard depreciation.



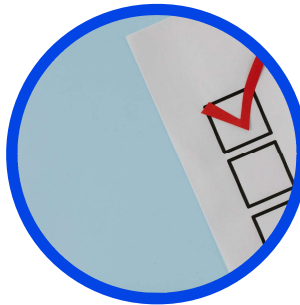
Feature Penetration & Trends

Assessing how common these features are across different vehicle "Eras."



Price Distribution & Variance

Visualizing the association between feature availability and price fluctuations.



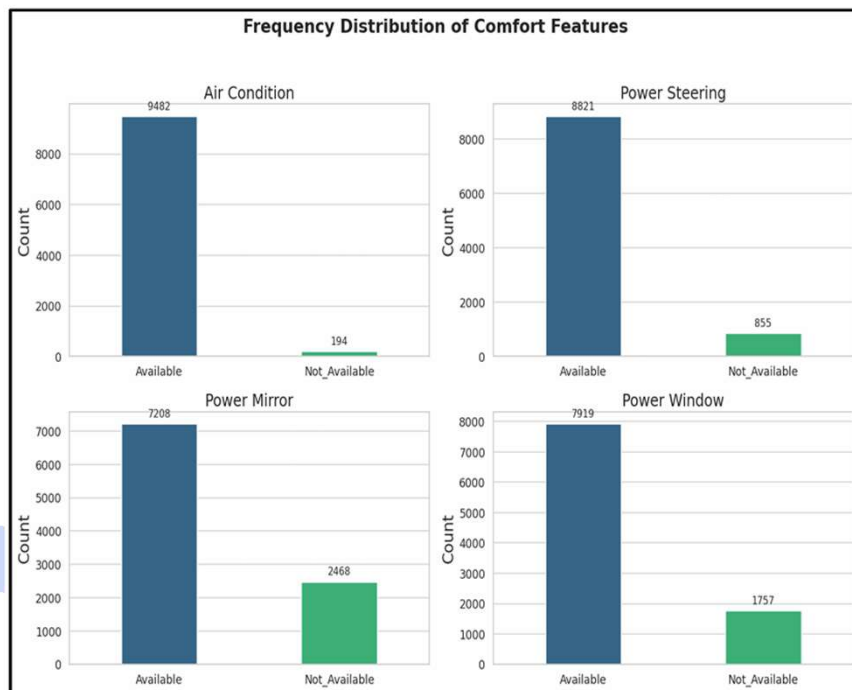
Statistical Validation

Using Cramér's V and non-parametric tests to assess the significance of feature availability and pricing.

OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Feature Availability Distribution (Count Plots)



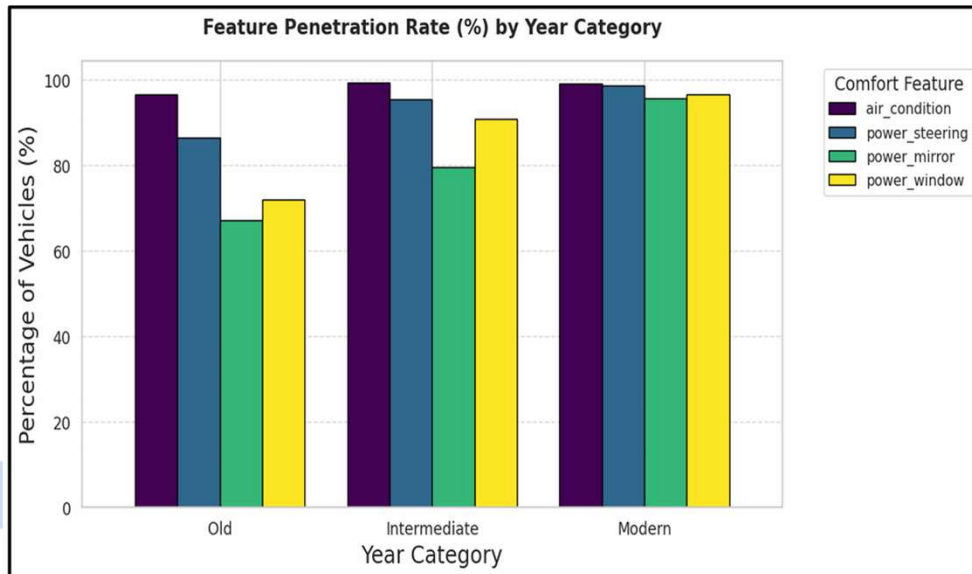
- Air Conditioning (AC) shows near-total market saturation, present in over 95% of vehicles.
- Power Steering, Power Windows, and Power Mirrors also exhibit high prevalence across listings.
- These features have transitioned from premium add-ons to baseline consumer expectations.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Feature Availability Rate Across Vehicle Eras (Clustered Column Chart)



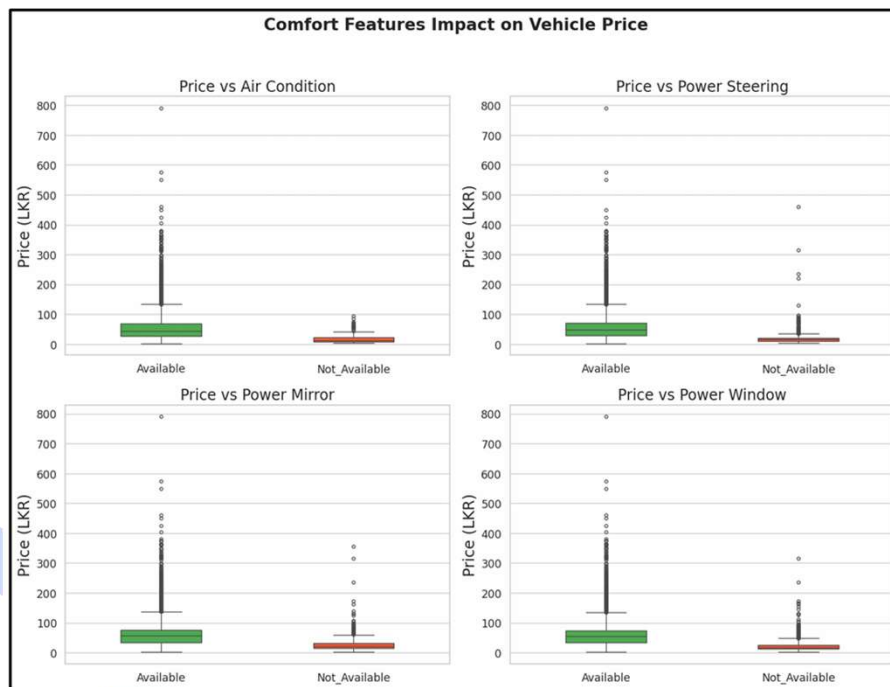
- Air Conditioning maintained consistently high availability rates across all manufacturing eras.
- Power Windows and Power Mirrors show noticeable availability growth during (Intermediate) 2010–2017.
- Modern vehicles demonstrate uniformly high availability rates, indicating feature standardization.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Price Distribution by Feature Availability (Box Plots)



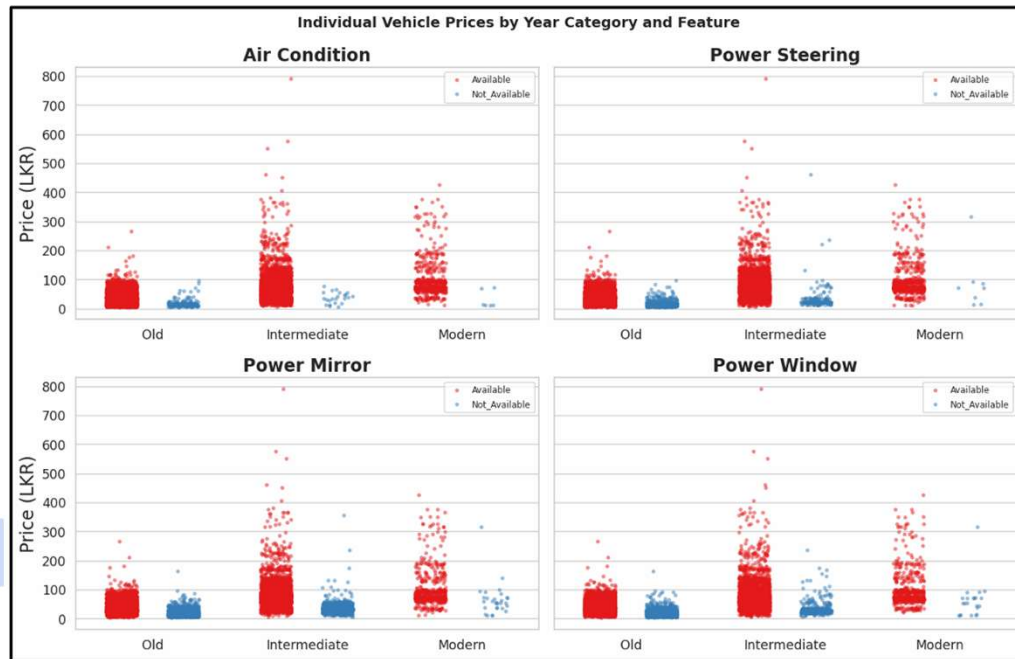
- Vehicles with comfort features consistently show higher median prices.
- Feature-equipped vehicles exhibit wider price ranges, including high-value outliers.
- Comfort features are strongly associated with higher-value and premium vehicle segments.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Individual Price Variation Across Vehicle Era and Feature Availability (Strip Plots)



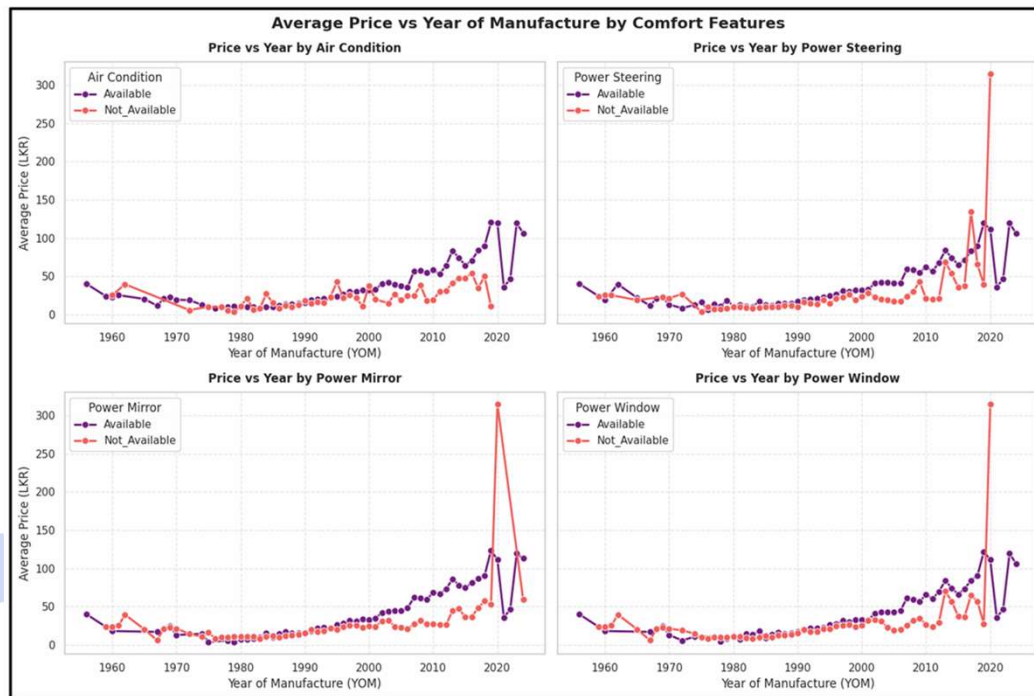
- Older vehicles show limited price variation regardless of comfort features.
- Intermediate and Modern vehicles display substantial price variability based on features.
- Comfort features act as significant value multipliers in newer vehicle categories.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Price Trends by Year of Manufacture and Feature Availability (Line Chart)



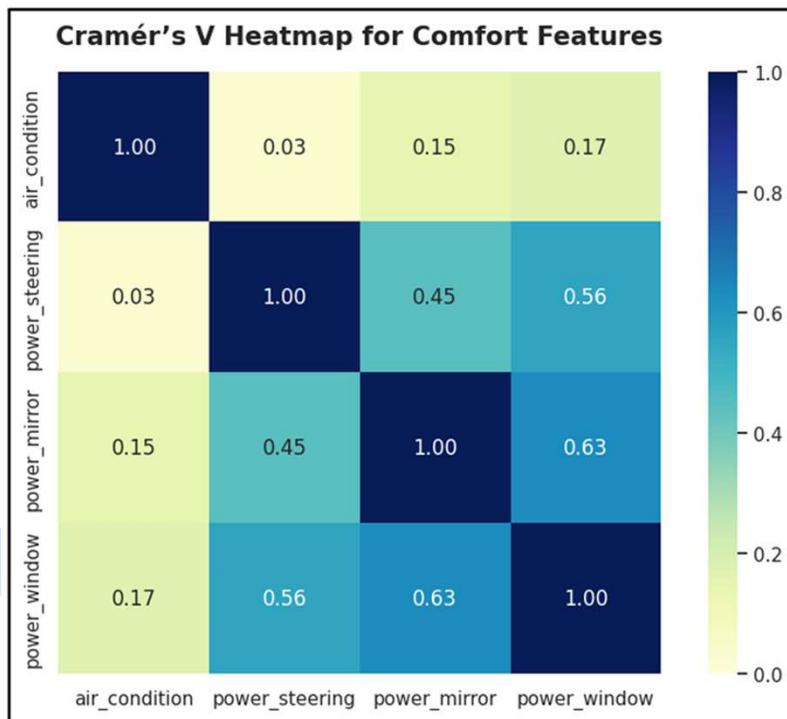
- Average vehicle prices increase steadily for vehicles manufactured after 2000.
- Sharp price spikes occur around 2020, particularly for vehicles lacking certain features.
- Recent years show increased price volatility, reflecting unstable market conditions.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Feature Association Analysis (Cramér's V Heatmap)



- Strong association exists between Power Windows and Power Mirrors ($V \approx 0.63$).
- These features are commonly bundled together in vehicles.
- AC shows weak correlation with other features, confirming its universal availability.



OBJECTIVE NO 3

EDA / BIVARIATE ANALYSIS

Does price significantly vary based on the availability of comfort features?

Test	Variable	Statistic	p-value	Conclusion
Normality (D'Agostino K ²)	Price	7,493.8353	0.0000e+00	Not Normal
Levene's Test	Air Condition	51.7698	6.7041e-13	Unequal Variances
Levene's Test	Power Steering	237.8043	5.0491e-53	Unequal Variances
Levene's Test	Power Mirror	522.5523	1.0996e-112	Unequal Variances
Levene's Test	Power Window	418.1072	5.1980e-91	Unequal Variances

Test	Variable	Statistic	p-value	Conclusion
Mann-Whitney U	Air Condition	1,552,860	1.0241e-60	Significant
Mann-Whitney U	Power Steering	6,651,322	1.3266e-298	Significant
Mann-Whitney U	Power Mirror	15,247,108	0.0000e+00	Significant
Mann-Whitney U	Power Window	12,250,961	0.0000e+00	Significant

H_0 : Median price is equal for vehicles with and without feature i .

H_1 : Median price differs between vehicles with and without feature i .

- D'Agostino K² test confirmed price distribution is non-normal ($p < 0.05$).

- Levene's test indicated unequal variances across comfort feature groups suggesting non-parametric tests.

- Mann-Whitney U test confirmed statistically significant price differences for all comfort features ($p < 0.05$).





Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

OBJECTIVE 04:



Market Sentiment and
Geo-Economic Analysis



OVERVIEW

- This Purpose of this objective is examines whether geographic location and financial accessibility through leasing influence vehicle market price in Sri Lanka's second-hand vehicle market.
- The analysis focuses on differences between Urban and Non-Urban markets and evaluates whether leasing availability creates price segmentation.
- This objective is important because understanding whether vehicle prices and availability vary by location and financing options helps reveal the level of market integration and accessibility within the national vehicle marketplace.



Market Accessibility

How geographic location (Urban vs Non-Urban) influences vehicle availability and market participation across regions.



Financial Accessibility

How leasing availability affects vehicle prices and whether it lowers entry barriers for buyers.



Market Integration

Whether pricing patterns remain consistent across locations, indicating a unified national market rather than regional segmentation.

OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Distribution of Listings by Location Type

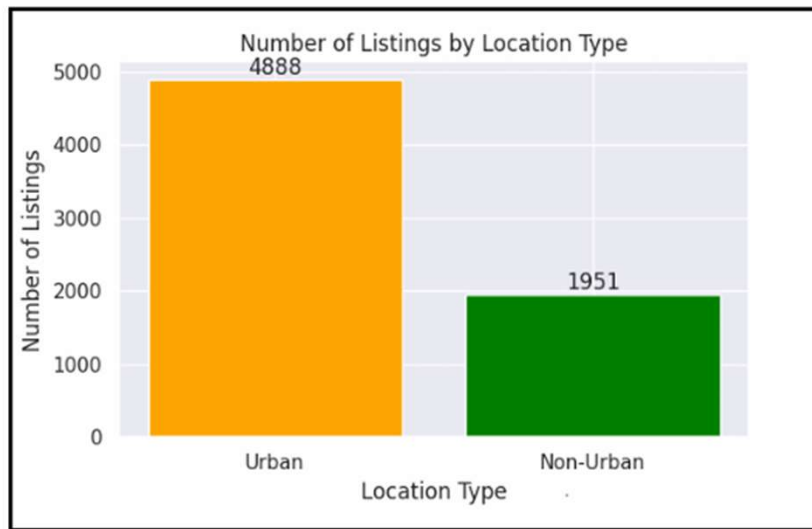


Figure 27: Distribution of vehicle listings by location type.

- Vehicle listings were classified into Urban and Non-Urban categories based on the administrative status of towns.
- Urban locations include Municipal and Urban Council areas, while Non-Urban locations include Pradeshiya Sabha and transitioning areas.
- vehicle listings are heavily concentrated in Urban locations, more than twice compared to non-urban areas, indicating that market activities are significantly higher in urban centers.



OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Distribution of Listings by Leasing Availability



Figure 28: Distribution of vehicle listings by leasing availability.

- The majority of vehicle listings are non-leasing vehicles, while only a small fraction of listings offer leasing options.
- This imbalance suggests that leasing is not a widespread financing mechanism within the market.
- Due to this imbalance, special care is required when interpreting price differences involving leasing availability.



OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Price Trends by Year of Manufacture Across Location Types

- Average vehicle prices show a strong upward trend with year of manufacture for both Urban and Non-Urban markets.
- The price trajectories for Urban and Non-Urban listings are almost identical across years, indicating similar pricing behavior.
- This similarity suggests that vehicle age, rather than geographic location, is a primary driver of vehicle prices.

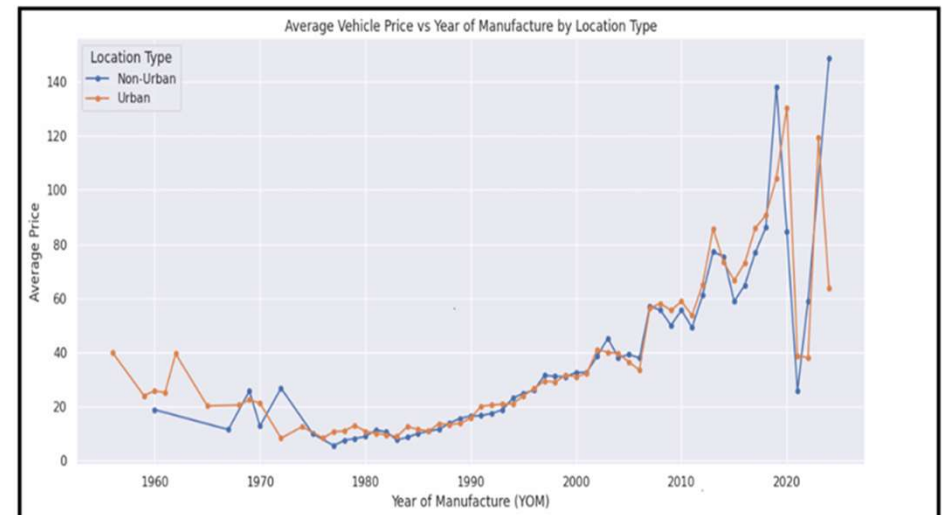


Figure 29: Average vehicle price by year of manufacture for Urban and Non-Urban markets.

OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Price Trends by Year of Manufacture and Leasing Availability

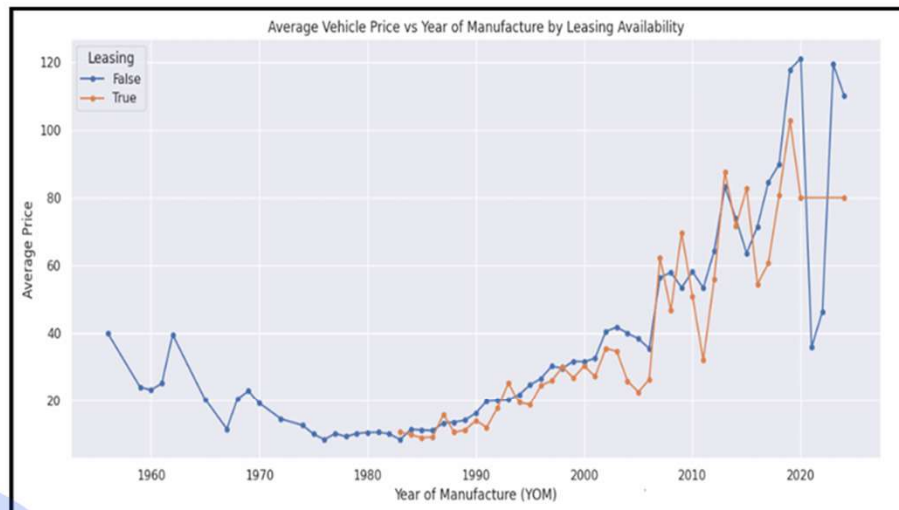


Figure 30: Average vehicle prices by year of manufacture, grouped by leasing availability.

- When prices are examined by year of manufacture and leasing availability, both groups exhibit the same upward trend with newer vehicles.
- Leasing and non-leasing price curves closely follow each other over time.
- This indicates that year of manufacture remains a dominant pricing factor, even when leasing availability is considered.



OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Geographic Concentration of Market Activity (Towns)

- The top fifteen towns account for a large share of total vehicle listings.
- Colombo dominates the market, followed by Gampaha, Kurunegala, and Kandy.
- This concentration highlights that vehicle market activity is spatially uneven, with major urban centers functioning as key market nodes.

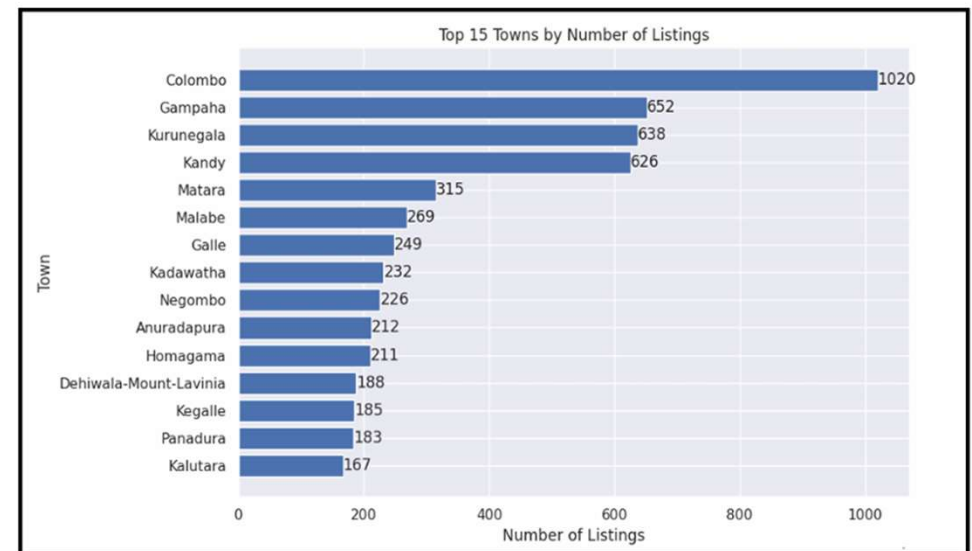


Figure 31: Top fifteen towns by number of vehicle listings.

OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Geographic Distribution by Province

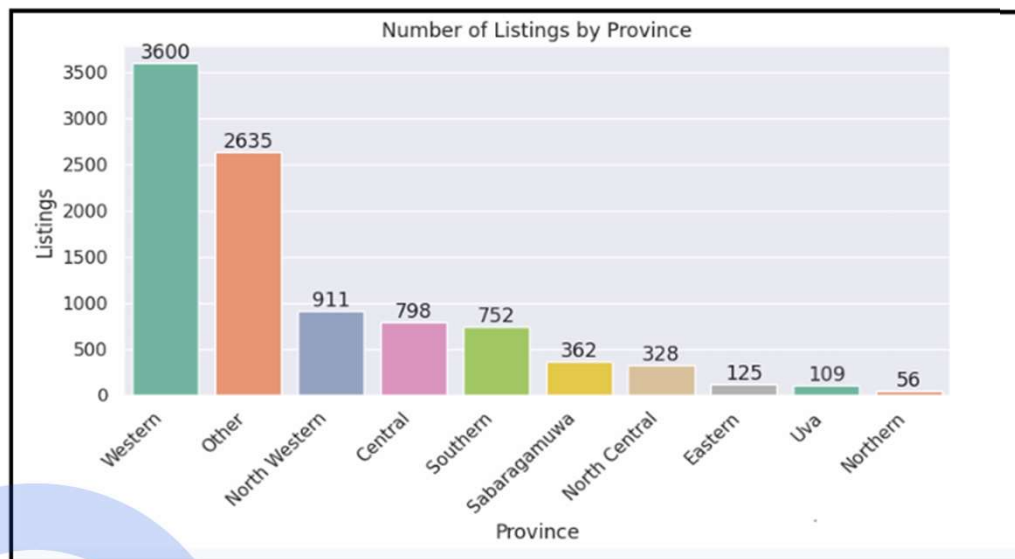


Figure 32: Distribution of vehicle listings by province.

- The Western Province accounts for the highest number of vehicle listings.
- Moderate listing volumes are observed in North Western, Central, and Southern provinces.
- Eastern, Uva, and Northern provinces contribute relatively small numbers of listings, indicating uneven regional market participation.

OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Method for Evaluating Price Differences

- To assess whether location type and leasing availability influence vehicle prices, a non-parametric bootstrap approach was used.
- This method was chosen because vehicle prices are highly skewed and contain extreme values, making parametric assumptions unsuitable. Additionally, the category levels for both location and leasing variables are imbalanced, particularly for leasing availability.
- The analysis focuses on the difference in median vehicle prices between two groups at a time (Urban vs Non-Urban, and Leasing vs No Leasing).
- For each comparison, prices within each group were resampled with replacement 10,000 times.
- The median price difference was calculated in each resample to generate a bootstrap distribution.
- A 95% bootstrap confidence interval was constructed from this distribution.
- Statistical significance was determined by checking whether the confidence interval includes zero.

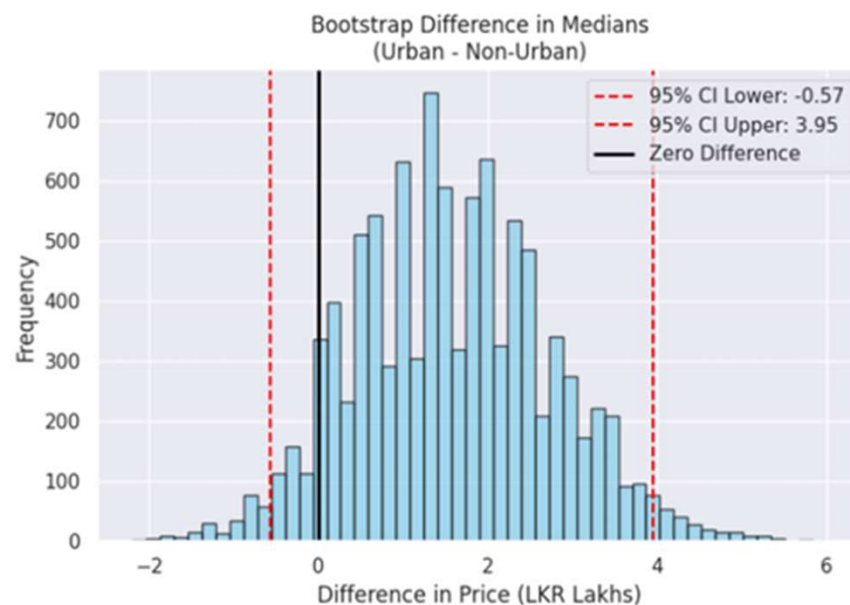
OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Location-Based Price Difference

$$H_0: \text{Median}_{(\text{Urban})} - \text{Median}_{(\text{Non-Urban})} = 0$$

$$H_1: \text{Median}_{(\text{Urban})} - \text{Median}_{(\text{Non-Urban})} \neq 0$$



95% Confidence Interval for Difference in Medians (Urban - Non-Urban): (-0.5746, 3.9500)
Mean Bootstrapped Difference: 1.5691

- The 95% confidence interval for the difference in median prices between Urban and Non-Urban listings includes zero.
- This indicates no statistically significant difference in median vehicle prices across location types.
- Although the average difference is slightly positive, it is not statistically meaningful.
- This suggests that the vehicle market is largely integrated across locations.

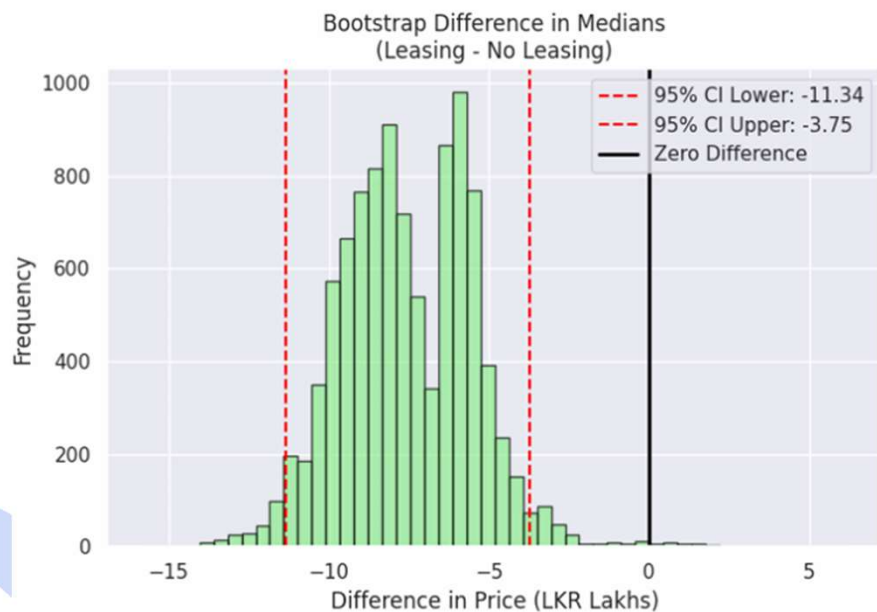
OBJECTIVE NO 4

EDA / BIVARIATE ANALYSIS

Leasing-Based Price Difference

$$H_0: \text{Median}_{(\text{Leasing})} - \text{Median}_{(\text{Leasing})} = 0$$

$$H_1: \text{Median}_{(\text{Leasing})} - \text{Median}_{(\text{Leasing})} \neq 0$$



95% Confidence Interval for Difference in Medians (Leasing - No Leasing): (-11.3375, -3.7500)
Mean Bootstrapped Difference: -7.5622

- The 95% bootstrap confidence interval for the difference in median prices between leasing and non-leasing vehicles does not include zero.
- The entire interval lies below zero, indicating a statistically significant price difference.
- Vehicles with leasing availability are typically priced lower than non-leasing vehicles.
- This suggests that leasing functions as a financial accessibility mechanism rather than a premium pricing feature.



Determinants of Vehicle (Car) Valuation in the Sri
Lankan Online Secondary Market

MODELING FOR INFERENCE



STABILIZING THE MODEL WITH LOG-TRANSFORMATIONS



The Log-Price Transformation ($\log(\text{price})$)

- Goal: Stabilizes variance and corrects the "right-skewness" typical of car prices (where a few luxury cars stretch the data).
- Result: Normalizes price distribution for better statistical inference.

The Log-Engine Capacity Transformation ($\log(\text{engine_cc})$)

- Goal: Captures Diminishing Marginal Effects.
- Logic: A 500cc increase matters significantly more for a small car than for a large SUV.
- Benefit: Allows for Elasticity-based Interpretation (e.g., "A 10% increase in engine size correlates to an X% increase in price").

VISUAL VS. STATISTICAL REALITY



While the model improved significantly after transformation, we performed a "Stress Test" comparing visual cues against formal statistical tests.

Assumption	Visual Diagnostic (The Look)
Normality	Q-Q Plot: Revealed departures from the diagonal line.

Key Takeaway

Visual inspections can be deceiving. The failure of these tests - despite the "cleaner" appearance of the plots - is exactly why we moved toward Non-Parametric methods and GAMs. It ensures that our conclusions about brand value aren't built on a flawed statistical foundation.

BEYOND LINEAR LIMITS: WHY A GAM?



When linear models (OLS) fail, it is usually because they are trying to force "straight-line" logic onto a "curved" reality.

- **The Limitation of OLS:**

- Assumption Failure: The violation of normality and homoscedasticity was a symptom of a deeper issue - market trends like depreciation are inherently non-linear.
- The "Straight Line" Trap: OLS treats every year of age or every cc of engine size with the same constant weight, which doesn't reflect how buyers actually behave.

- **The GAM Advantage: Flexibility + Interpretability**

- Capturing Real-World Trends: GAMs are designed to map complex patterns, such as rapid early-life depreciation and the late-life price plateaus common in the second-hand market.
- Additive Structure: Unlike "Black-Box" machine learning models (like Random Forests), GAM allows us to isolate and visualize the specific impact of each feature (Age, Brand, Engine, etc...) on the final price.

The "Best of Both Worlds" Solution

More Accurate than OLS: It adapts to the data's natural "wiggleness" rather than forcing it into a straight line.

More Transparent than AI: It provides clear, actionable curves that stakeholders can understand at a glance.

GAM RESULTS: HIGH EXPLANATORY POWER



The Generalized Additive Model successfully captured the nuances that the linear model missed, providing a highly reliable fit for the Sri Lankan market.

- **Explaining the Variance:**

- Pseudo R^2 : 0.85
- The model explains 85% of the variation in vehicle prices, indicating a high degree of reliability for statistical inference.

- **Confirming Non-Linearity (EDoF):**

- Both Age and $\log(\text{engine_cc})$ showed an Effective Degrees of Freedom (EDoF) of approximately 14.
- **What this means:** Since the EDoF is much higher than 1 (linear), it statistically proves that these relationships are highly curved and non-linear, fully justifying the move away from OLS.

- **The "Vintage" Reversal:**

- The age curve shows standard depreciation for decades, but for vehicles older than 50 years, the price curve begins to rise.
- **Insight:** This marks the transition from "used car" to "vintage/collector asset."

SIGNIFICANT DRIVERS VS. MARKET NOISE



While engine and age are the primary drivers, the model also evaluated the "value add" of specific features and location.

1. Statistically Significant Drivers

Beyond engine and age, these features significantly move the needle on price:

- Brand
- Gear (Manual/Automatic)
- Fuel Type (Diesel/Petrol/Hybrid)
- Leasing

2. Non-Significant & Counter-Intuitive Findings

- Location (Province): Price differences between provinces were minor and unstable, suggesting a unified national online market where location doesn't dictate price.
- Power Windows: Surprisingly found to be statistically non-significant.
- The Data "Bias" Warning: Features like A/C and Power Steering showed counter-intuitive effects.
 - Diagnosis: Wide confidence intervals in the plots suggest this is due to data sparsity or reporting bias (e.g., almost all cars in the dataset have A/C, so the "No A/C" sample is too small to be reliable).



Determinants of Vehicle (Car) Valuation in the Sri Lankan
Online Secondary Market

FINAL TAKEAWAYS (CONCLUSIONS)



THE DYNAMICS OF THE SRI LANKAN MARKET



The second-hand vehicle market is not driven by a single factor, but by a complex interplay of brand, mechanics, and market psychology.

- **Market Heterogeneity:**

- The market is defined by extreme diversity—from budget Suzuki hatchbacks to luxury imports.
- Toyota acts as the psychological "middle-market" anchor, defining value retention for the entire country.

- **The Premium Drivers:**

- Alternative Fuels: Hybrid and Electric vehicles command a clear price premium.
- Transmission: Automatic transmissions consistently price higher than manuals, reflecting a shift in consumer preference.
- Bundled Features: Comfort features (A/C, Power Steering) add the most value when "bundled," especially during periods of vehicle scarcity.

- **National Integration:**

- The data confirms a unified national market; location does not significantly dictate price, but financial accessibility (leasing options) does.

FINAL TAKEAWAYS



- **The Analytical Triumph:**

- Moving from OLS to GAM was essential. The GAM captured the "curves" of the market that linear models ignored, successfully explaining 85% of price variations.

- **Key Market Lessons:**

- Brand is a Financial Shield: Toyota and Honda provide the best protection against depreciation.
- Age isn't Linear: Depreciation is aggressive early on but reverses for "Vintage" assets (50+ years).
- Engine Size is a Proxy: engine_cc tells us the vehicle's "class" (Budget vs. Luxury) more than just its power.

Final Verdict:

Second-hand pricing in Sri Lanka is a rational response to brand reputation, mechanical performance, and scarcity, where high-demand brands and modern features consistently sustain price premiums.



Determinants of Vehicle (Car) Valuation in the Sri
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**Determinants of Vehicle (Car) Valuation in the
Sri Lankan Online Secondary Market**

THANK YOU

FOR YOUR ATTENTION



